



جامعة أبوظبي
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Assignment 3

CSC408: Distributed Information Systems

**Implementing a Spring Boot-based Secure RESTful API-
Based Peer-to-Peer Skills Exchange Platform – SkillSwap**

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Introduction

The CSC408 course project involves creating the peer-to-peer skill-sharing portal, SkillSwap UAE. It promotes connections, knowledge sharing, and skill sharing, all of which are important components of the UAE's 2025 "Year of Community" project. The objective is to establish a supportive and welcoming online community where people may teach and learn many talents, including languages, culinary arts, coding, and more. Users can sign up for the platform as mentors, learners, or both. Learners can ask to join sessions, and mentors can post their talents. By taking part, users receive credits that may be redeemed for prizes or access to more advanced classes. Utilizing an emphasis on security, scalability, and user-friendly RESTful APIs, the system is constructed utilizing Spring Boot.

This report describes the design and construction of the system, including the database architecture, user authentication, incentive system, API structure, testing, and suggestions for future enhancements.

System overview

MySQL is used for the database and Spring Boot for the backend development of the SkillSwap UAE platform. The Eclipse IDE is used to build the project, which has a straightforward Model-View-Controller (MVC) structure. The user interface for pages such as dashboard, registration, and login is created using custom HTML and CSS files.

Technologies Used:

- Spring Boot: which manages backend functionality and builds the RESTful API.
- MySQL: for storing users, skills, session data, credits, and feedback.
- HTML/CSS: for creating frontend pages.
- Eclipse: as the development environment to run the program.

Key Features:

- User registration and login with roles (mentor, learner, or both).

- Mentors can create and publish skills.
- Learners can view and register for available skills.
- A credit/reward system based on skill participation.
- A shared dashboard that shows all published skills and each user's registered skills.

Characteristics of the Web App

Users can learn and share skills through the SkillSwap web application. It is constructed with plain HTML/CSS for the front end and Spring Boot for the back end. Users may sign up as learners, mentors, or both. Learners can sign up for the skills that mentors publish. The backend has a standard MVC structure and stores data in MySQL.

The dashboard, profile, rewards, login, and registration are all done through static HTML pages. To communicate with the system, the backend makes REST API endpoints available.

Security Implemented

The Passwords of users is being hashed using BCrypt therefore the passwords are securely stored in the database. JWT utility class is implemented for token generation & reading but it was not fully integrated. Therefore, the API endpoints are not really protected by login or roles. Spring Security is configured but it does not check each request

Areas for Improvement

- JWT tokens are not being used properly after user loges in
- Anyone can gain access to any API without being logged in and more security filters should be implemented.
- User input isn't checked very well
- If a learner registers to a skill, he can just select complete and claim the points without any confirmation or approvals
- Front-end design is simple and may not be compatible with most devices
- No email is used to check when someone register therefore anyone could sign up without confirmation

Summary

It's a good starting point for people to exchange skills and learn new skills. Strong technologies used like spring boot and MySQL make it advanced in the way it works, but some parts of the web app need to be more secured. Proper login checks, improved user interface and functioning credit system would make the web app way better

References

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- c. *Spring boot :: Spring boot. (n.d.). <https://docs.spring.io/spring-boot/docs/current/reference/html/>*

